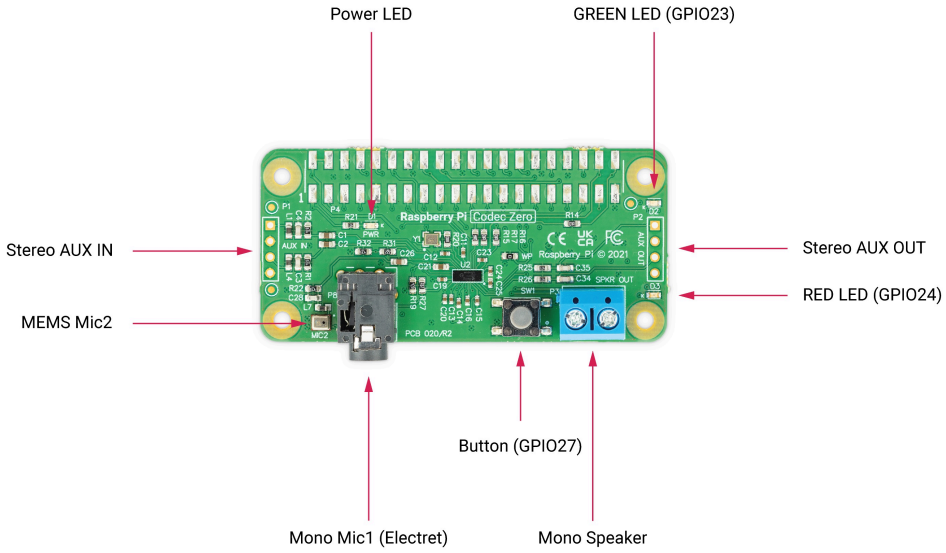
Raspberry Pi Codec Zero

Source: https://github.com/raspberrypi/documentation/blob/7376533ab86ea07acc77ef6a9f3c3e57fcff833d/documentation/asciidoc/accessories/audio/codec_zero.adoc (CC BY-SA 4.0; adapted for this benchmark)
Image source:
https://raw.githubusercontent.com/raspberrypi/documentation/7376533ab86ea07acc77ef6a9f3c3e57fcff833d/documentation/asciidoc/accessories/audio/images/Codec_Zero_Board_Diagram.jpg (CC BY-SA 4.0)



Raspberry Pi Codec Zero

Raspberry Pi Codec Zero

Raspberry Pi Codec Zero is a Raspberry Pi Zero-sized audio HAT. It delivers bi-directional digital audio signals (I2S) between a Raspberry Pi and the Codec Zero's on-board Dialog Semiconductor DA7212 codec. The Codec Zero supports a range of input and output devices.

High performance 24-bit audio codec

Supports common audio sample rates between 8-96 kHz

Built in micro-electro-mechanical (MEMS) microphone (Mic2)

Mono electret microphone (Mic2 left)

Automatic MEMS disabling on Mic2 insert detect

Supports additional (no fit) mono electret microphone (Mic1 right)

Stereo auxiliary input channel (AUX IN) – PHONO/RCA connectors

Stereo auxiliary output channel (Headphone/AUX OUT)

Flexible analogue and digital mixing paths

Digital signal processors (DSP) for automatic level control (ALC)

Five-band EQ

Mono line-out/mini speaker driver: 1.2 W at 5 V, THD < 10%, R = 8 Ω

The Codec Zero includes an EEPROM which can be used for auto-configuration of the Linux environment if necessary. It has an integrated MEMS microphone, and can be used with stereo microphone input through a 3.5 mm socket and a mono speaker (1.2W/8Ω).

In addition to the green (GPIO23) and red (GPIO24) LEDs, a tactile programmable button (GPIO27) is also provided.

Pinouts

P1/2 | Support external PHONO/RCA sockets if needed. P1: AUX IN, P2: AUX OUT.

P1 | Pin 1 is square.

Codec Zero is an ideal design starting point for small-scale projects such as walkie-talkies, smart doorbells,

vintage radio hacks, or smart speakers.